

# Joshua Himmens

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Engineering Physics Student

## Technical Skills

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**Machine Learning** | TensorFlow, Keras, PointNet, Weights and Biases (wandb), ONNX

**Embedded Programming** | Experienced with FreeRTOS on TMS570, RP2040, and STM32, KiCAD

**Software Development** | Python, Rust, C, C++, Java, MATLAB, Bash, Git

**Quantum Computing** | Qiskit, QASM, Pasqal, QCNNs, VQCs

**Particle Physics** | ROOT, CERN grid computing, Athena

## Work Experience

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### Associate Machine Learning Developer

AltaML | Jan 2025 – Apr 2025 | [himmens.com/altaml](http://himmens.com/altaml)

- Developed machine learning models using scikit-learn to save the Government of Alberta >\$4 million annually.
- Developed novel evaluation metrics to minimize regional bias and maximize transferability of models.
- Built Azure ML pipelines for automated model training and deployment, serving predictions via REST APIs integrated with SQL databases and Blob Storage.
- Implemented Uber's H3 hexagonal spatial indexing to balance geographic accuracy with user privacy in an infrastructure planning tool.

### ATLAS Deep Learning Research Student

TRIUMF | May 2024 – Feb 2025 | [himmens.com/triumf](http://himmens.com/triumf)

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting-edge transfer learning approaches.
- Implemented models into production using NVIDIA Triton.

### Command and Data Handling (CDH) Lead and Firmware Developer

UBC Orbit Satellite Design Team | Oct 2023 – Present | [himmens.com/orbit](http://himmens.com/orbit)

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of 10 firmware developers, with over \$15 000 worth of technical infrastructure.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

### Embedded Firmware Developer

UBC Orbit Satellite Design Team | Jan 2023 – Present | [himmens.com/orbit](http://himmens.com/orbit)

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computing system using FreeRTOS on a TMS570 microcontroller.

## Publications and Presentations

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**Developing Machine Learning Techniques for Particle Flow in the ATLAS Experiment** at

Canadian Astroparticle Physics Summer Student Talks Competition (CASST 2024)

Jun 2024

- Placed 2nd of 44 presentations.

**JetPointNet: A Machine Learning Approach to Cell-to-Track Attribution in the ATLAS**

**Experiment** at Canadian Undergraduate Physics Conference

Oct 2024

**3D Particle Flow in the ATLAS Calorimeter: How to Train Your Model** at TRIUMF Science  
Week May 2024  
**ALEASAT ESA "Fly Your Satellite!" Training Week Presentation** at European Space Agency's  
ESEC-GALAXIA (Transinne, Belgium) Mar 2024

## Leadership and Advocacy

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### Chair

UBC IEEE Aerospace Electrical  
Systems Student Branch Chapter  
Aug 2025 - Present

- Served as \*Technical Program Chair\* for the Student Aerospace Summit, September 2025.
- Organized events to unify aerospace-interested students across UBC.

### Highly Qualified Personnel (HQP) Advisory Committee Member

Arthur B McDonald Astroparticle  
Physics Institute  
Jan 2024 - Sep 2024

- Worked with members across Canada to develop opportunity lists for students and recent graduates.
- Shared McDonald Institute opportunities with eligible HQP in BC and Alberta.

### Advisory Team Member

Child Rights Connect  
Jan 2021 - Dec 2023

- Provided guidance to UN delegations on communication strategies for high-level rights goals.
- Presented to governments and consulted on international initiatives to support the UN Convention on the Rights of the Child.

## Additional Experience

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- **"Quantum School for Young Students"** Participant in intensive quantum computing program
- **"Introduction to Quantum Computing"** Participant in 8-month course using IBM's quantum infrastructure
- **"Scientific Computing with Python certification"** 300 hours of training
- **"GENQ Global Hackathon (Geneva, 2025 and Calgary, 2024)"** Build QCNNs and QUBO solvers for quantum machine learning tasks

## Honors and Awards

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### Hector J. MacLeod Award

Apr 2024

Awarded to students who have demonstrated pioneering efforts in their chosen field of study and attained high scholastic honours.

### Erich Vogt First Year Summer Research Experience (FYSRE) Award

May 2024

Awarded to promising students in physics for a 4-month research placement.

### Alberta Centennial Award and Alberta Premier's Citizenship Award

May 2023

Awarded for outstanding community service.

### Calgary Flames Foundation Community Involvement Scholarship

Aug 2023

Awarded for community involvement.

### Julia Turnbull Leadership Award for exceptional community service

Jul 2023

Awarded for exceptional community service.